

- 5-9 Consider application of the conservation principles of Chapter 2 to control volume $A-B-C-D$ for the steady laminar boundary-layer problem sketched in Fig. 5P-9.

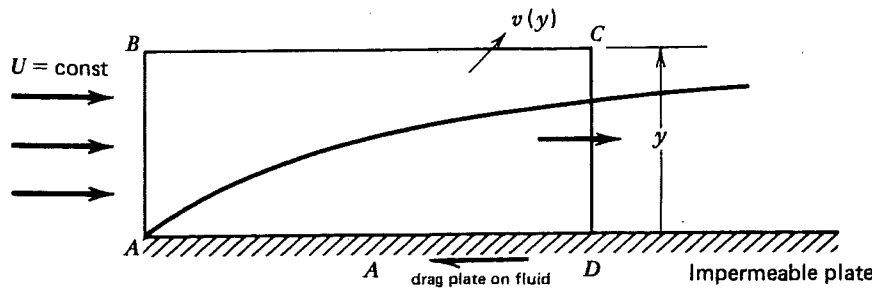


Figure 5P-9

- (a) Show that the mass flow rate across $B-C$ equals

$$\int_0^y (\rho U - \rho u) dy = \int_{x_A}^{x_D} \rho v(y) dx$$

- (b) Show that the drag force exerted on the fluid by the plate is

$$\int_0^y \frac{\rho U U}{g_c} dy - \left[\int_0^y \frac{\rho u u}{g_c} dy + \int_{x_A}^{x_D} \frac{\rho v(y) U}{g_c} dx \right] - F_{\text{drag plate on fluid}} = 0$$

- (c) Use the results of parts a and b to show that, when $y \rightarrow \infty$ to avoid shear stresses that exist near the plate,

$$F_{\text{drag}} = \frac{\rho}{g_c} \int_0^\infty u(U - u) dy$$

and

$$\delta_2 = x \frac{\bar{C}_f}{2} = 0.66412 \left(\frac{\nu x}{U} \right)^{1/2}$$

- 5-13 Consider steady parallel flow of a constant-property fluid over a flat plate. When $\text{Pr} \rightarrow 0$, the thermal boundary layer is much thicker than the velocity boundary layer, so that throughout the thermal boundary layer $F \approx \eta$ since then $u/U = F' \approx 1$. Such is the case for liquid metals.

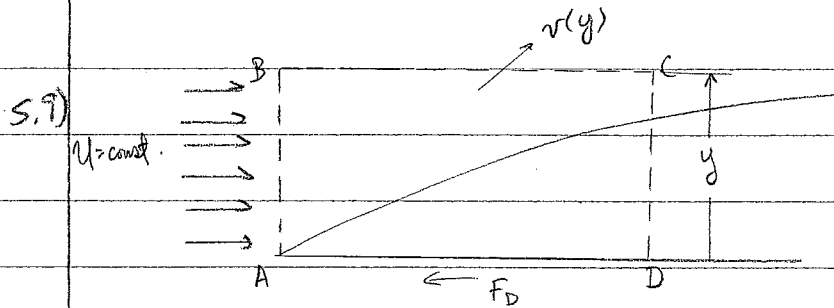
- (a) Show that in such a case Eq. (5-26) gives

$$\frac{d\theta(0)}{d\eta} = \left(\frac{\text{Pr}}{\pi} \right)^{1/2}$$

- (b) Show that the result of part a gives, with $\text{Nu} = hx/k$ and $\text{Re} = Ux/\nu$,

$$\lim_{\text{Pr} \rightarrow 0} \frac{\text{Nu}}{\text{Re Pr}^{1/2}} = \frac{1}{0.33206 \pi^{1/2}} \frac{C_f}{2} = 1.7 \frac{C_f}{2}$$

Chpt 5 / 9.13



Steady laminar boundary-layer problem

S.9A) The mass flow rate across BC is

$$\sum \dot{m}_{IN} = \sum \dot{m}_{OUT}$$

$$\int_{AB} \rho U dy = \int_{CD} \rho u dy + \int_{BC} \rho v dx$$

$$\left| \int_0^y (\rho U - \rho u) dy = \int_{x_A}^{x_B} \rho v(y) dx \right|$$

S.9B) The drag force exerted on the fluid by the plate is found by summing up forces in x-dir.

$$\sum F_x = \left\{ \text{rate of momentum change in CV} + \text{rate of momentum flow across CV} \right\} \quad \text{O (S.S.)}$$

$$(\text{rate of momentum out}) - (\text{rate of momentum in}) = F_D$$

$$\left[-\frac{1}{\rho c} \int_0^y \rho U u dy + \frac{1}{\rho c} \left(\int_{x_A}^{x_B} \rho U v(y) dx + \int_0^y \rho u u dy \right) \right] = -F_D$$

5.9c) From part a, for constant properties,

$$\int_0^y \left(1 - \frac{u}{u_\infty}\right) dy = \int_{x_A}^{x_B} \frac{v}{u_\infty} dx \quad \text{--- (1)}$$

From part b,

$$g_c \frac{F_D}{\rho u_\infty^2} = \int_0^y dy - \int_0^y \frac{u^2}{u_\infty^2} dy - \int_{x_A}^{x_B} \frac{v}{u_\infty} dx$$

$$g_c \frac{F_D}{\rho u_\infty^2} = \int_0^y \left(1 - \frac{u^2}{u_\infty^2}\right) dy - \int_{x_A}^{x_B} \frac{v}{u_\infty} dx$$

Using eqn (1)

$$\begin{aligned} g_c \frac{F_D}{\rho u_\infty^2} &= \int_0^y \left(1 - \frac{u^2}{u_\infty^2}\right) dy - \int_0^y \left(1 - \frac{u}{u_\infty}\right) dy \\ &= \int_0^y \left(\frac{u}{u_\infty} - \frac{u^2}{u_\infty^2}\right) dy \end{aligned}$$

$$g_c \frac{F_D}{\rho u_\infty^2} = \int_0^y \frac{u}{u_\infty} \left(1 - \frac{u}{u_\infty}\right) dy \quad \text{--- (2)}$$

$$F_D = \frac{\rho}{g_c} \int_0^y u(u - u_\infty) dy$$

As $y \rightarrow \infty$,

$$F_D = \frac{\rho}{g_c} \int_0^\infty u(u - u_\infty) dy$$

Since the momentum thickness is defined as

$$\delta_2 = \frac{1}{U^2} \int_0^{\infty} u(U-u) dy$$

$$\delta_2 = \int_0^{\infty} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$$

which is of the same form as eqn (2). Therefore

$$g_c \frac{F_D}{\rho U^2} = \delta_2 \quad \text{--- (3)}$$

The drag force is defined on p. 170 as

$$F_D = \frac{1}{2} \rho U^2 \bar{C}_f x$$

Inserting this expression into eqn (3) & let $g_c = 1$,

$$\frac{\frac{1}{2} \rho U^2 \bar{C}_f x}{\rho U^2} = \delta_2$$

$$\delta_2 = \frac{1}{2} x \bar{C}_f$$

Since (p. 170)

$$\bar{C}_f = 2 C_f = 2 (0.332) \sqrt{\frac{\nu}{Ux}}$$

we get

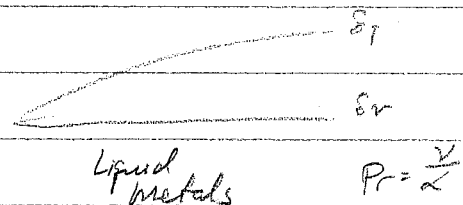
$$\boxed{\delta_2 = 0.694 \sqrt{\frac{\nu x}{U}}}$$

S.13) Parallel flow over flat plate

Constant property

When $Pr \rightarrow 0$, then the thermal BL

becomes mostly $F \approx \eta$ & $\frac{\eta}{x} = F' \approx 1$



S.13a) Equation 5.26 is given by

$$\theta'' + \frac{Pr}{2} F \theta' = 0, \quad \theta(0) = 0, \quad \theta(\infty) = 1$$

Since $F \approx \eta$,

$$\theta'' + \frac{Pr}{2} \eta \theta' = 0$$

The integrating factor for $\theta'' + a \theta' = 0$ is

$$\sigma = e^{\int \frac{Pr}{2} \eta d\eta} = e^{\frac{Pr}{4} \eta^2}$$

The ODE becomes

$$(\theta' e^{\frac{Pr}{4} \eta^2})' = 0$$

$$\theta' e^{\frac{Pr}{4} \eta^2} = C_1$$

$$\theta' = C_1 e^{-\frac{Pr}{4} \eta^2} \quad \text{--- (1)}$$

Integrating again,

$$\theta = C_1 \int_0^\eta e^{-\frac{Pr}{4}s^2} ds + C_2$$

Let $r^2 = \frac{Pr}{4}s^2$

$$r = \frac{\sqrt{Pr}}{2}s, \quad dr = \frac{\sqrt{Pr}}{4} ds$$

$$s = \frac{2}{\sqrt{Pr}} r, \quad ds = \frac{2}{\sqrt{Pr}} dr$$

Limits: $\eta_{old} = \frac{2}{\sqrt{Pr}} \eta_{new}$

$$\eta_{new} = \frac{\sqrt{Pr}}{2} \eta_{old}$$

$$\theta = C_1 \frac{2}{\sqrt{Pr}} \int_0^{\frac{\sqrt{Pr}}{2}\eta} e^{-r^2} dr + C_2$$

The error function is defined as

$$\text{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-x^2} dx, \quad \begin{matrix} \text{erf}(0)=0 \\ \text{erf}(\infty)=1 \end{matrix}$$

The solution, using erf, becomes

$$\theta = C_1 \frac{2}{\sqrt{Pr}} \frac{\sqrt{\pi}}{2} \text{erf}\left(\frac{\sqrt{Pr}}{2}\eta\right) + C_2$$

Define $C_3 = C_1 \frac{\sqrt{\pi}}{\sqrt{Pr}}$,

$$\theta = C_3 \text{erf}\left[\frac{\sqrt{Pr}}{2}\eta\right] + C_2$$

Evaluating at BC's,

$$\theta(0) = 0 = C_3 \operatorname{erf}(0) + C_2 \Rightarrow C_2 = 0$$

$$\theta(\infty) = 1 = C_3 \operatorname{erf}(\infty) \Rightarrow C_3 = 1$$

The solution becomes

$$\theta(\eta) = \operatorname{erf}\left[\frac{\sqrt{Pr}}{2}\eta\right]$$

The derivative of $\theta(\eta)$ is given by equation 0 as

$$\theta' = C_1 e^{-\frac{Pr}{4}\eta^2}$$

Evaluating at $\eta=0$ gives

$$\left.\frac{d\theta}{d\eta}\right|_{\eta=0} = C_1 e^{(0)}$$

$$\theta'(0) = C_1$$

Since C_1 was previously defined as

$$C_3 = C_1 \sqrt{\frac{\pi}{Pr}}$$

$$C_1 = \sqrt{\frac{Pr}{\pi}} C_3$$

We have

$$\theta'(0) = \sqrt{\frac{Pr}{\pi}}$$

5.13b) With $Nu = \frac{h_x}{k}$ and $Re = \frac{u_x}{\nu}$, the result of (a) with

$$\theta = \frac{T - T_\infty}{T_w - T_\infty} \rightarrow T = \theta(T_w - T_\infty) + T_\infty$$

$$\eta = y \sqrt{\frac{u}{\nu x}} \rightarrow y = \eta \sqrt{\frac{\nu x}{u}}$$

We evaluate the heat flux at the wall

$$q_w = -k \frac{\partial T}{\partial y} \Big|_{y=0}$$

$$= -k \frac{(T_w - T_\infty)}{\sqrt{\frac{\nu x}{u}}} \frac{\partial \theta}{\partial \eta} \Big|_{\eta=0}$$

$$q_w = -k \sqrt{\frac{u}{\nu x}} (T_w - T_\infty) \theta'(0) \quad \text{--- (2)}$$

The convective HT is defined as

$$q_w \equiv h (T_w - T_\infty) \quad \text{--- (3)}$$

Equating ② and ③:

$$-k \sqrt{\frac{u}{\nu x}} (T_w - T_\infty) \theta'(0) = h (T_w - T_\infty)$$

$$h = k \sqrt{\frac{u}{\nu x}} \sqrt{\frac{Pr}{\pi}}$$

The Nusselt Number is

$$Nu_x = \frac{hx}{k}$$

$$= \frac{x}{k} \left(k \sqrt{\frac{u}{\nu x}} \sqrt{\frac{Pr}{\pi}} \right)$$

$$Nu_x = Re_x^{1/2} \sqrt{\frac{Pr}{\pi}} \quad \text{--- ④}$$

where $Re_x = \frac{u x}{\nu}$.

Rearranging equation ④ into form given in the book,

$$\frac{Nu}{Re Pr^{1/2}} = \frac{1}{\sqrt{\pi} Re}$$

Since the drag coefficient is defined as (p.170)

$$C_f = \frac{0.664}{\sqrt{Re_x}}$$

$$\frac{Nu_x}{Re_x Pr^{1/2}} = \frac{1}{\sqrt{\pi}} \frac{C_f}{0.664}$$

Since part a solves for $Pr \rightarrow 0$,

$$\lim_{Pr \rightarrow 0} \frac{Nu}{Re Pr^{1/2}} = \frac{1}{\sqrt{\pi}} \frac{C_f}{0.664}$$